

Zero Waste Food Management

Project Overview

The Local Food Wastage Management System is a Streamlit-based web application developed to help reduce food wastage by connecting food providers with receivers through a centralized platform. The system enables surplus food listings, claim management, user registration, and data-driven insights to support efficient food redistribution within communities.

Key Objectives

- Create a platform for sharing surplus food instead of wasting it.
 - Connect food providers and receivers through a structured workflow.
 - Enable secure management of food listings and claims.
 - Provide visibility into food redistribution activities through analytics.
 - Maintain a simple and accessible solution using Python, Streamlit, and SQLite.
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System Design and Functionality

The application is built using Streamlit for the user interface, Python for backend logic, and SQLite as the database.

The platform begins with a basic login system where users create a username and password to access the application. Once logged in, users can register themselves as food providers or food receivers depending on their role.

Core Features

User Management

- User registration and login.
- Provider registration.
- Receiver registration.
- User-specific access controls.

Food Listing Management

Providers can:

- Add new food listings.
- View their own listings.
- Update existing listings.
- Delete listings.

Claim Management

Receivers can:

- Browse available food listings.
- Submit claims for food items.
- Track claim status.

Providers can:

- Review incoming claims.
- Approve or deny requests made against their food listings.

CRUD Operations

The system supports complete Create, Read, Update, and Delete (CRUD) functionality across registered users, food listings, and claims. Users can only modify records that belong to their own accounts, ensuring controlled access and data integrity.

Database Structure

The SQLite database stores:

- User information
- Provider records
- Receiver records
- Food listings
- Claim records

Relationships between tables are maintained through primary and foreign key constraints to ensure consistency and reliability of data.

Food Claim Workflow

1. A provider lists surplus food.
 2. A receiver submits a claim for the listed item.
 3. The provider reviews the request.
 4. The provider either approves or denies the claim.
 5. The claim status is updated and reflected throughout the system.
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Analytics

The application includes an analytics dashboard that provides insights into food redistribution activities, provider participation, receiver engagement, listing trends, and claim statistics through interactive SQL-powered visualizations.

To support comparison and analysis, users can switch between:

- Original dataset (static database)
- Live application data (active database)

This allows users to compare the initial data with changes generated through platform usage.

User Interface Design

The application follows a single-page design with a sidebar navigation menu and custom styling.

Understand

Provides information about the project mission, food wastage challenges, and platform impact metrics.

Analytics

Contains interactive dashboards, charts, and reports generated from SQL queries and visualized using Plotly.

Register

Allows users to register as providers or receivers through structured forms.

Act

Serves as the operational section where providers manage food listings and receivers submit claims, while approved users can review and process requests.

Technology Stack

- Python
- Streamlit

- SQLite
- Pandas
- SQL
- Plotly
- Custom CSS Styling